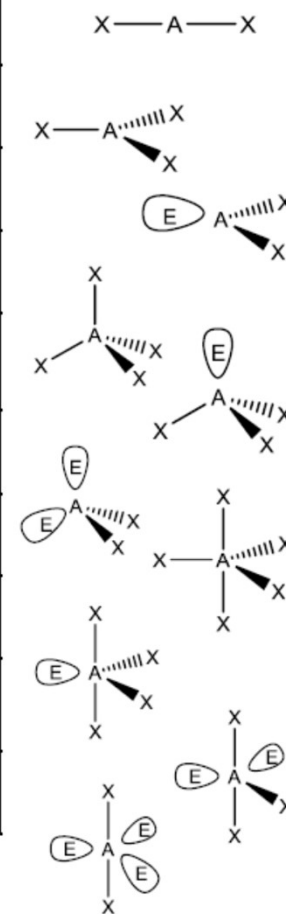
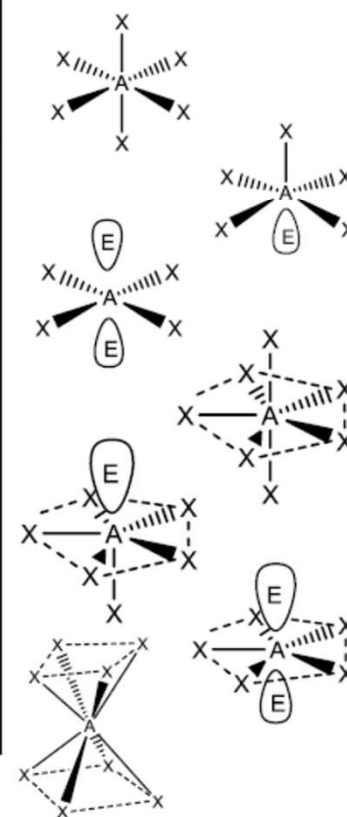


Arrangement	Bonding pairs X	Lone pairs E	Occupancy/ Steric number	Shape of the molecule	Examples
Linear	2	0	2	Linear AX_2	CO_2 , CS_2 , HCN , $BeCl_2$, N_2O , NO_2^+ , NCF , $NCBr$
Triangular	3	0	3	Triangular planar AX_3	SO_3 , BF_3 , $COCl_2$, NO_3^- , CO_3^{2-} , BO_3^{3-}
	2	1	3	Bent (V shape) AX_2E	SO_2 , O_3 , $NOCl$, $SnCl_2$, PbF_2 , NOF
Tetrahedral	4	0	4	Tetrahedral AX_4	CH_4 , $SiCl_4$, SO_2Cl_2 , $POCl_3$, XeO_4 , NSF_3 , ClO_4^- , PO_4^{3-}
	3	1	4	Triangular Pyramidal AX_3E	NH_3 , PCl_3 , $SOCl_2$, IO_3^-
	2	2	4	Bent AX_2E_2	H_2O , SCl_2 , NSF , ICl_2^+ , $SeCl_2$, $[I(C_6H_5)_2]^+$, KrF_3^+
Trigonal bipyramid	5	0	5	Trigonal Bipyramidal AX_5	PF_5 , SF_4O
	4	1	5	See saw AX_4E	SF_4 , XeO_2F_2 , IF_4^+ , IOF_3 , $IO_2F_2^-$
	3	2	5	Bent T shape AX_3E_2	ClF_3 , BrF_3 , $XeOF_2$, $I(Ph)[(CF_3C(O)O)]_2$
	2	3	5	Linear AX_2E_3	XeF_2 , I_3^- , ICl_2^- , BrF_2^- , $(C_6F_5)_2Xe$



Arrangement	Bonding pairs X	Lone pairs E	Occupancy / Steric number	Shape of the molecule	Examples
Octahedral	6	0	6	Octahedral AX_6	SF_6 , PF_6^- , SiF_6^{2-} , AlF_6^{3-} , IF_6^+
	5	1	6	Square pyramidal AX_5E	BrF_5 , XeF_5^+ , IOF_4^- , TeF_5^- , SbF_5^{2-} , $[C_6H_4C(O)O]I(OAc)_3$
	4	2	6	Square planar AX_4E_2	XeF_4 , IF_4^- , $XeOF_3^-$
Pentagonal bipyramid	7	0	7	Pentagonal bipyramid AX_7	IF_7 , $IO_2F_5^{2-}$, IOF_6^- , TeF_7^- , $CH_3OTeF_6^-$, $(CH_3O)_2TeF_5^-$, BiF_7^{2-} , SbF_7^{2-}
	6	1	7	Pentagonal Pyramid AX_6E	$XeOF_5^-$, IOF_5^{2-} , $[IF_5(C_6H_5)]^-$, XeF_6 (-monocapped octahedron)
	5	2	7	Pentagonal planar AX_5E_2	XeF_5^- , IF_5^{2-}
Square antiprism	8	0	0	Square antiprism AX_8	IF_8^- , XeF_8 , XeF_8^{2-} (AX_8E)



6.3. SOME PROBLEMS

Problem I. : Which one of CH_3^- and CF_3^+ is pyramidal and which one is planar?

CH_3^- has $4 + 3 + 1$ (negative charge) = 8 electrons i.e. 4 electron pairs. Of these three are bond pairs and one is a lone pair. Hence this has to be pyramidal. It is a carbanion.

CF_3^+ has $4 + 3 - 1$ (positive charge) = 6 electrons around carbon. So there are three bond pairs and hence the shape has to be planar. This is a carbonium ion.

Problem II. : Predict the shape of thionyl dichloride.

Thionyl dichloride is SOCl_2 . Sulphur being less electronegative than oxygen or chlorine serves as the central atom. The chlorines being one electron short of an octet are linked to sulphur by single bonds. Oxygen being two electrons short will be linked either by (a) a double bond or (b) a coordinate link. If (a) is the case then the valence shell of sulphur has

$6 + 1 + 1 + 2 = 10$ electrons *i.e.* five pairs. Of these one makes a pi bond which is innocent in respect of the shape of a molecule. So we are left with three bond pairs and a lone pair



Fig. 6.12. Lewis representation of SOCl_2

so that the shape is pyramidal (cf : NH_3). If (b) is the case then the valence shell of sulphur has $6 + 1 + 1 = 8$ electrons *i.e.* four pairs. Of these three are bond pairs and one is a lone pair. Once again the predicted shape is pyramidal.

Problem III : Predict the shape of sulphuryl (sulphonyl) dichloride.

Sulphuryl dichloride is SO_2Cl_2 . The central element sulphur is bonded to the two chlorines via single bonds and to the two oxygens via double bonds. Then valence shell of sulphur has $6 + 1 + 1 + 2 + 2 = 12$ electrons. Of the six electron pairs the two pi bond pairs will not influence the shape of the molecule. So we are left with four electron pairs which are all bond pairs. A tetrahedral structure follows.

Alternatively we may assume that the two oxygens are linked via coordinate bonds from sulphur. Then the valence shell of sulphur has $6 + 1 + 1 = 8$ electrons. Thus there are four bond pairs leading to a tetrahedral shape for the compound.



Fig. 6.13. Lewis representation of SO_2Cl_2

Problem IV : Predict the shape Of SnCl_3^- .

Here tin acts as the central element. Chlorine being one electron short of an octet will be linked to tin by single bond. The number of electrons around tin in its valence shell is given by $4 + 1 + 1 + 1 + 1$ (negative charge) = 8. These electrons are present as three bond pairs and one lone pair. So the shape is pyramidal (a tetrahedron with a lone pair).

Problem VI : Predict the shape of arsonium ion.

Arsonium ion is $[\text{AsH}_4]^+$. All the four hydrogens will be linked to arsenic via single bonds. Valence shell of arsenic has $5 + 1 + 1 + 1 + 1 - 1$ (positive charge) = 8 electrons *i.e.* four pairs. All these are bond pairs. Therefore predicted shape is tetrahedral (cf : NH_4^+ , PH_4^+).

Problem VII : Predict the shape of the borohydride ion.

Borohydride ion is $[\text{BH}_4]^-$, boron being the central element. All the four hydrogens are linked to boron through single bonds. Valence shell electrons around boron number $3 + 1$

+ 1 + 1 + 1 + 1 (negative charge) = 8 electrons *i.e.* four pairs. All of these are bond pairs. There is no lone pair. So the shape is tetrahedral.

Problem VII : Predict the shape of HCN molecule.

Carbon acts as the central element. Lewis structure shows a single bond between carbon and hydrogen and a triple bond (one sigma and two pi) between carbon and nitrogen. The valence shell of carbon has $4 + 1 + 3 = 8$ electrons.

So we have four bond pairs of which two are pi and two are sigma bonds. The two sigma bonds predict a linear molecule. The pi bonds have no influence on the ultimate shape of the molecule.

Problem VIII : Predict the shape of nitrosyl chloride.

Nitrosyl chloride is ClNO. The central element nitrogen is linked to chlorine through a single bond and to oxygen via a double bond. Hence the valence shell of nitrogen has $5 + 1 + 2 = 8$ electrons. There are three bond pairs and one lone pair. The pi bond between nitrogen and oxygen does not influence the shape of the molecule. Thus the two sigma bond pairs and the lone pair should roughly be at 120° . Since lone pair repels bond pairs more strongly the bond angle $\angle \text{ClNO}$ should be less than 120° . In fact the bond angle is about 116° . It is V-shaped.

Problem IX : Predict the shape of chlorate ion.

Chlorate ion is ClO_3^- . The central element chlorine is bonded to the three oxygen atoms through coordinate covalent bonds. The valence shell of chlorine is made up of $7 + 1$ (negative charge) = 8 electrons *i.e.* four electron pairs. Of these four pairs three are bond pairs and one lone pair. Therefore the molecule will have a pyramidal shape like NH_3 .

Problem X. : Predict the shape of carbonium ion.

A carbonium ion is a group of atoms having a carbon atom with only six electrons around it and is formed as a result of, say heterolytic cleavage of a carbon halogen bond. The carbon electron goes over to the halogen making a halide ion X^- and a positive carbonium ion like $(\text{CH}_3)_3\text{C}^+$. The valence shell of the central carbon has $4 + 3 - 1$ (positive charge) = 6 electrons, which make the C—C bond pairs. It follows that the three bond pairs will attain maximum stability when they are at 120° in a plane *i.e.* the shape will be a flat triangular planar one.

On the hybridisation model the central carbon is assumed to be sp^2 hybridised, with the third (say p_z) orbital being empty. The three unpaired electrons in the three sp^2 hybrid orbitals then will overlap with suitable orbitals having electrons with opposite spin. Once again the shape turns out to be a triangular planar one.

Problem XI. : Predict the shape of carbanion.

A carbanion is an anion (CH_3^-) with negatively charged carbon. It has three shared pair covalent bonds and a lone pair. The shape will then be pyramidal (cf. NH_3), one of the three vertices of a tetrahedron being occupied by the lone pair.

On the hybridisation model carbon is sp^3 hybridised.

Problem XII. : Explain the T-shape of ClF_3 .

The valence shell of the central element chlorine has $7 + 1 + 1 + 1 = 10$ electrons *i.e.* five electron pairs. Of these three are bond pairs and two lone pairs. A trigonal bipyramidal arrangement is expected. According to rule 1 of VSEPR the two lone pairs should take up

the two axial positions. But it is an empirical fact that lone pairs prefer to take up equatorial positions. Lone pairs are fatter than bond pairs and this fact probably prompts the lone pairs to occupy equatorial positions. Thus in ClF_3 , two lone pairs and one fluorine take up the equatorial plane while the other two fluorines occupy axial sites. In such an environment each equatorial lone pair feels two 90° repulsion and one 120° repulsion with the bond pairs. If the lone pairs occupied axial positions then each had to face three 90° repulsion with the three equatorial bond pairs. Experimental structure determination reveals that a lone pair prefers two 90° repulsions and one 120° repulsion. However this explanation is not beyond criticism.

BrF_3 has a shape like that of ClF_3 .

Problem XIII. : Explain the irregular tetrahedral shape of the POCl_3 molecule with Cl-P-Cl bond angle being 103° .

The POCl_3 molecule [Fig. 6.9(a)] has a double bond with oxygen and three single bonds to the three chlorines. Valence shell electrons around phosphorus number $5 + 2 + 1 + 1 + 1 = 10$. These ten electrons are present in three single bonds and one double bond (a σ + a π bond). Since the pi bond does not influence the shape, the latter will be decided by the four electron pairs. So the molecule will be tetrahedral. But the fat $\text{P}=\text{O}$ double bond will exert greater repulsive force on the single $\text{P}-\text{Cl}$ bonds than do the $\text{P}-\text{Cl}$ bonds on one another. Hence the Cl-P-Cl bond angles are lowered below the ideal tetrahedral angle 109° while the Cl-P-O angle is raised above 109° .

Problem XIV. : PCl_5 has the shape of a trigonal bipyramid whereas IF_5 has the shape of a square pyramid. Account for this difference.

The number of valence shell electrons around phosphorus in PCl_5 is $5 + 1 + 1 + 1 + 1 + 1 = 10$, the five chlorine atoms being linked to phosphorus by single bonds. These ten electrons will be present in five shared pair covalent bonds. VSEPR theory predicts a TBP structure. On the other hand the valence shell electrons of iodine in IF_5 number $7 + 1 + 1 + 1 + 1 + 1 = 12$, there being five shared pair covalent bonds and one unshared lone pair. Ideal arrangement for six pairs of electrons will be an octahedron. IF_5 therefore will have the five F atoms towards five corners of an octahedron while the sixth corner will be occupied by the unshared lone pair. So effectively IF_5 molecule will be square pyramidal.

Problem XV. : Predict the shape of F_2O .

Here oxygen acts as the central element. The two fluorines are connected to oxygen by single bonds. Therefore the valence shell of oxygen has $6 + 1 + 1 = 8$ electrons i.e. four pairs. Of these two are bond pairs and two are lone pairs. Hence the molecule is V-shaped.

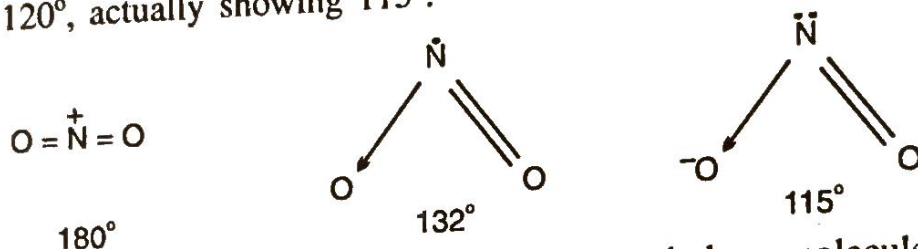
Problem : Explain the bond angles in the series NO_2 (180°), N^+O_2 (132°) and NO_2^- (115°).

The structure of N^+O_2 involves a coordinate link from nitrogen to one oxygen and a double bond to the other oxygen. This means that nitrogen has a total of $5 + 2 - 1$ (positive charge) = 6 electrons i.e. three pairs. Of these three pairs one is a pi bond while the other two are sigma. Pi bond does not influence the shape of a molecule or ion. So in order to minimise repulsion the two sigma bonds should ideally be at 180° i.e. the ion should be linear. Alternatively and more simply we may assume the two oxygens to be double bonded to nitrogen. Each double bond would then be made up of one sigma bond and a pi bond. The valence shell of nitrogen will then have $5 + 2 + 2 - 1 = 8$ electrons. Since the pi bonds

do not influence the shape the two sigma bonds will prefer to stay at 180° in order to minimise repulsion thus giving a linear geometry.

In neutral NO_2 we cannot but assume a valence shell of $5 + 0 + 2 = 7$ electrons around nitrogen. (If we had assumed double bonds to the two oxygens then valence shell of nitrogen would have $5 + 2 + 2 = 9$ electrons, which is beyond an octet—the limit of electrons for elements of the short period). This would mean three electron pairs plus a lone electron. Of the three electron pairs one is a coordinate link to one oxygen and the other two make a double bond (sigma and pi) to the other oxygen. Two σ bond pairs ideally should give a linear geometry. However the lone electron has some repulsive effect on the other two pairs and hence the bond angle is much less than 180° —in fact it is 132° .

In NO_2^- the negative sign i.e. an extra electron is assumed to be located on one oxygen by virtue of the greater electronegativity of oxygen. The valence shell of nitrogen is made up of $5 + 1 + 2 = 8$ electrons. This means four electron pairs: a single shared pair bond with the mononegative oxygen, a double bond to the other oxygen and a lone pair. Leaving aside the pi bond we have three electron pairs: two bond pairs and one lone pair. Because of strong repulsion by the lone pair the angle between the two bond pairs further shrinks from the ideal 120° , actually showing 115° .



Problem XVII : Predict the shapes of acetylene and ethylene molecules.

Acetylene is C_2H_2 . Lewis representation allows an octet for each carbon if we assume a triple bond between the two carbons and a single bond to each hydrogen. Then each carbon has $4 + 1 + (1 + 1 + 1) = 8$ electrons i.e. 4 electron pairs of which two are pi bonds which do not influence the shape of a molecule. Then we are left with two σ bond pairs only, which means that the molecule has to be linear.

Ethylene is C_2H_4 . Lewis representation gives two hydrogens bonded to each carbon by single bonds and the two carbons linked by a double bond. Then each carbon has $4 + (1 + 1) + (1 + 1) = 8$ electrons i.e. 4 bond pairs. Of these one is a pi bond. We are then left with three σ bond pairs on each carbon, which gives each of the two CH_2 units a triangular planar shape.

Problem XVIII. : Predict the shape of ClO_2^- ion.

Assuming for simplicity sake that the extra negative charge comes on to chlorine, giving it an octet, then two coordinate links to the two oxygens will give them octets. Then there are two lone pairs and two bond pairs on chlorine. So a tetrahedral disposition of four electron pairs and a V-shape for the ion are predicted (cf : H_2O).

Problem XIX. : Predict the shapes of carbonyl chloride and carbonyl sulphide.

Carbonyl chloride is COCl_2 . Carbon, being less electronegative, will serve as the central element. The two chlorines are linked to carbon by single bonds and the oxygen by a double bond (one sigma and one pi). Carbon therefore has in its valence shell $4 + 1 + 1 + 2 = 8$ electrons i.e. 4 electron pairs. Of these four, one is a pi bond pair, which does not, influence

the shape of a molecule. We are thus left with three σ bond pairs which, according to VSEPR model will give a triangular planar molecule.

Carbonyl sulphide is COS. Both oxygen and sulphur are linked to carbon by double bonds (sigma and pi). Carbon has in its valence shell $4 + 2 + 2 = 8$ electrons i.e 4 electron pairs. So we have only two σ electron pairs which will determine the shape of the molecule. VSEPR model predicts a linear molecule (cf : CO₂).

Problem XX. : How the oxonium ion will look like?

The oxonium ion is H₃O⁺. The central element oxygen has three single bonds with the three hydrogens. Valence shell of oxygen has $6 + (1 + 1 + 1) - 1$ (due to the positive charge) = 8 electrons i.e 4 pairs of electrons. Evidently there are three bond pairs and one lone pair, giving a pyramidal shape to the ion (cf: NH₃). One of the four vertices of the tetrahedron is occupied by the lone pair.

STUDY QUESTIONS